



Clinical Bridge

Final Project Report

Bridging the Clinical Context Gap: An LLM-Powered Multi-Agent System for Synthesizing Electronic Health Records, Remote Patient Monitoring, and Anamnesis Data

Field	Value
Course	COP-3442: Prompt Engineering
Institution	Bahcesehir University - Artificial Intelligence Engineering Department
Project Type	Context-aware multi-agent clinical decision-support prototype
Date	June 2026
Repository	github.com/lunahhammo/clinicalbridge
Students	Luna Hammo (2104910); Celina Joudi (2104885)

Safety Notice: ClinicalBridge is an educational proof-of-concept using entirely simulated data. It is not a clinical tool and must not be used for real patient care.

Table of Contents

1. Executive Summary.....	4
2. Problem Statement: The Clinical Context Gap.....	4
2.1 Why This Is a Prompt Engineering Problem	5
3. Project Objective and Scope	5
3.1 Primary Objective	5
3.2 Scope of the Project.....	6
3.3 Project Limitations.....	6
4. Proposed Solution Architecture.....	6
4.1 Technology Stack.....	7
<i>The technology stack was selected to provide a lightweight and modular architecture while supporting reliable prompt execution, semantic retrieval, and structured output generation. Lang Chain was used to coordinate interactions between agents, Chroma DB enabled efficient retrieval of EHR documents through vector search, and GPT-4o provided consistent structured responses required for the Clinical Context Brief generation.....</i>	
5. Module-to-Capstone Mapping	8
6. Detailed System Components	9
6.1 Clinical Context Brief Output.....	10
7. Data Model and Simulated Dataset	10
7.1 Scenario Coverage.....	11
8. Prompt Engineering Specifications.....	12
8.1 Prompt Iteration Summary.....	13
9. Evaluation Framework and Results	13
9.1 Agent-Level Metrics.....	14
<i>The table presents the v2 evaluation results because they served as the baseline used to identify weaknesses during prompt iteration. The improvements introduced in v3 are summarized in the following sections and reflected in the final end-to-end evaluation.</i>	
9.2 End-to-End Metrics	15
9.3 Failure Analysis and Improvements.....	15
10. Project Milestones and Timeline.....	16
11. Deliverables Checklist	17
12. Ethical Considerations and Limitations.....	17

12.1 Critical Disclaimers.....	17
12.2 Ethical Design Principles Implemented.....	18
12.3 Known Limitations.....	18
12.4 Conclusion.....	19
<i>13. References.....</i>	<i>19</i>
13.1 Clinical Context Gap and Health Informatics	19
13.2 Remote Patient Monitoring.....	19
13.3 LLM Applications and Prompt Engineering	19
<i>Appendix A. Evidence and Verification Summary</i>	<i>20</i>

1. Executive Summary

ClinicalBridge is a context-aware, multi-agent clinical decision-support prototype developed for the COP-3442 Prompt Engineering capstone course. The system addresses the clinical context gap by integrating three fragmented healthcare data sources: Electronic Health Records (EHR), Remote Patient Monitoring (RPM), and structured patient history intake (anamnesis). Given a simulated RPM alert, ClinicalBridge generates a structured Clinical Context Brief (CCB) that combines current patient measurements with relevant medical history and patient-reported information to provide clinicians with a comprehensive overview of the patient's condition.

The implemented prototype consists of four specialized LLM-powered agents coordinated by a central orchestrator: the Alert Triage Agent, EHR Retrieval Agent, Anamnesis Agent, and Synthesis Agent. Each agent is designed with a dedicated prompt strategy, structured output format, temperature configuration, and safety constraints to ensure reliable and explainable behavior. This modular architecture demonstrates how prompt engineering extends beyond prompt writing to the design of coordinated, task-specific intelligent agents.

The prototype operates entirely on simulated clinical data, including 15 fictional adult patients, Electronic Health Records, Remote Patient Monitoring time-series data, structured anamnesis records, and five representative clinical evaluation scenarios. The final evaluation achieved successful completion of all five scenarios and passed all 40 scenario validation checks. Earlier evaluation results are also presented transparently, demonstrating progressive improvements through iterative prompt refinement, including a reduction in hallucinations to 3.2%, 91% source traceability, 100% safety compliance, and an average time-to-brief generation of 23.4 seconds.

Overall, ClinicalBridge demonstrates a practical, safety-aware proof of concept that uses prompt engineering, Retrieval-Augmented Generation (RAG), and multi-agent orchestration to synthesize fragmented clinical information into a structured, source-grounded Clinical Context Brief while preserving uncertainty and maintaining a human-in-the-loop approach to clinical decision support.

2. Problem Statement: The Clinical Context Gap

Healthcare professionals have access to large amounts of patient information, but much of it is stored across separate systems that are not easily connected. Electronic Health Records (EHRs) contain medical history, diagnoses, medications, allergies, laboratory results, and clinical notes. Remote Patient Monitoring (RPM) devices continuously collect physiological measurements such as blood pressure, glucose levels, or heart rate, while anamnesis provides patient-reported information including symptoms, medication adherence, lifestyle habits, and recent health concerns. Although each source provides valuable information, reviewing them together during a busy clinical workflow can be difficult and time-consuming.

This becomes especially challenging when an RPM alert is generated. Before deciding whether the alert requires immediate attention, clinicians often need to review the patient's medical

history, current medications, recent laboratory results, allergies, and self-reported symptoms. Since this information is typically distributed across multiple systems, gathering the necessary context takes time and increases the risk of overlooking important details. As a result, clinically significant changes may be missed, while less important alerts may receive unnecessary attention.

2.1 Why This Is a Prompt Engineering Problem

The challenge addressed by ClinicalBridge is not simply retrieving patient information, but combining information from multiple sources into a clear and reliable clinical summary. Since EHR records, RPM data, and anamnesis are stored in different formats, the system must accurately retrieve relevant information, identify relationships between data sources, and present only evidence that is supported by the available records. This makes prompt engineering a key part of the project, as the quality of each prompt directly influences the accuracy, consistency, and safety of the generated Clinical Context Brief.

Each agent was designed to solve a specific task within the workflow:

- **EHR Retrieval Agent:** Extracts clinically relevant information from structured and unstructured medical records while avoiding unsupported or fabricated facts.
- **Alert Triage Agent:** Evaluates incoming RPM measurements, identifies their level of urgency, and generates targeted queries for the remaining agents.
- **Anamnesis Agent:** Interprets patient-reported information and converts it into structured clinical observations without making assumptions beyond the provided data.
- **Synthesis Agent:** Combines evidence from all available sources into a single Clinical Context Brief, highlights conflicting or missing information, provides source-backed findings, and clearly communicates uncertainty whenever necessary.

3. Project Objective and Scope

3.1 Primary Objective

The primary objective of ClinicalBridge is to develop a context-aware, multi-agent clinical decision-support prototype capable of transforming a simulated Remote Patient Monitoring (RPM) alert into a structured Clinical Context Brief (CCB). To achieve this, the system retrieves and combines relevant information from Electronic Health Records (EHR), RPM data, and patient anamnesis using specialized LLM agents coordinated through a central orchestrator.

The generated Clinical Context Brief is intended to provide clinicians with the most relevant patient information in a concise and easy-to-review format. Rather than replacing clinical decision-making, the system is designed to support healthcare professionals by reducing the time required to gather context, highlighting important findings, and clearly communicating uncertainty when available information is incomplete or conflicting.

3.2 Scope of the Project

The ClinicalBridge prototype includes the following components:

- A functional Python-based proof of concept with a modular architecture and a single application entry point.
- Four specialized LLM agents coordinated by a central orchestrator responsible for workflow management and safety enforcement.
- A simulated clinical dataset consisting of Electronic Health Records (EHR), Remote Patient Monitoring (RPM) data, and structured anamnesis records.
- A prompt engineering portfolio documenting prompt design decisions, iterative improvements, and prompt versions throughout development.
- An evaluation framework that measures system performance using scenario-based, agent-level, and safety-focused metrics.
- Supporting documentation and a demonstration notebook that showcase the implementation and evaluation of the complete system.

3.3 Project Limitations

- Production clinical deployment, regulatory approval, or use with real patient records.
- Replacing physician judgment or issuing treatment orders.
- A full hospital EHR integration or live RPM device integration.
- Clinical validation against real-world outcomes.

4. Proposed Solution Architecture

ClinicalBridge follows a modular multi-agent architecture in which each agent is responsible for a specific stage of the clinical workflow. The system begins when an RPM alert is received, containing the patient's vital measurements and alert details. The orchestrator validates the incoming data and forwards it to the Alert Triage Agent, which determines the alert priority and generates retrieval queries for the remaining agents.

Based on the triage results, the orchestrator coordinates the EHR Retrieval Agent and the Anamnesis Agent to retrieve relevant medical history and patient-reported information in parallel. The outputs from both agents are then combined with the original RPM alert and passed to the Synthesis Agent, which generates the final Clinical Context Brief (CCB). Throughout the workflow, the orchestrator manages communication between agents, validates outputs, and applies safety rules to ensure reliable and consistent responses. If a critical alert is detected, the system can immediately recommend clinician review without waiting for the complete synthesis process.

Step	Component	Purpose
1	RPM Alert Input	Patient ID, timestamp, measurement type, value, baseline, threshold, trend context.
2	Alert Triage Agent	Classifies urgency and creates targeted EHR/anamnesis query parameters.
3	Orchestrator	Routes requests, handles critical bypass, coordinates parallel retrieval, validates outputs.
4	EHR Retrieval Agent	Retrieves relevant clinical information using Retrieval-Augmented Generation (RAG), identifies missing or outdated records, and provides traceable source references.
5	Anamnesis Agent	Extracts structured patient-reported information, including symptoms, medication adherence, lifestyle factors, over-the-counter medications, and patient concerns.
6	Synthesis Agent	Combines evidence from all agents to generate a structured Clinical Context Brief with source references, uncertainty reporting, and recommended clinician review actions.

4.1 Technology Stack

Layer	Implementation
Language/runtime	Python 3.12
LLM	GPT-4o, selected for its reliable structured JSON generation, instruction following, and LangChain integration
Embeddings	text-embedding-3-small
Vector store	ChromaDB local persistent store

Layer	Implementation
Orchestration	LangChain workflows, asynchronous Python execution ,explicit schema validation
Testing	pytest unit tests and a mock/live evaluation harness

The technology stack was selected to provide a lightweight and modular architecture while supporting reliable prompt execution, semantic retrieval, and structured output generation. Lang Chain was used to coordinate interactions between agents, Chroma DB enabled efficient retrieval of EHR documents through vector search, and GPT-4o provided consistent structured responses required for the Clinical Context Brief generation.

5. Module-to-Capstone Mapping

Course Module	ClinicalBridge Implementation
M1: Intro to LLMs	Model selection, comparison of available LLMs, and agent-specific temperature tuning to balance determinism and flexibility.
M2: Designing LLM Applications	Designed the overall multi-agent architecture, defined the clinician persona, Clinical Context Brief (CCB) format, and agent workflow.
M3: Prompt Content	Developed system prompts, few-shot examples, structured JSON output schemas, delimiters, and prompt constraints for each agent.
M4: Conversational Agency	Implemented the Anamnesis Agent to interpret patient-reported information while preserving context, intent, and uncertainty.
M5: Retrieval-Augmented Generation	Built the ChromaDB-based retrieval pipeline with semantic search, EHR chunking, source referencing, and missing-data detection.
M6: Evaluation	Evaluated the system using predefined clinical scenarios, gold-standard Clinical Context Briefs, agent-level metrics, and end-to-end performance analysis.

Course Module	ClinicalBridge Implementation
M7: Safety and Guardrails	Applied prompt guardrails including no-diagnosis rules, mandatory disclaimers, confidence scoring, allergy checks, and hallucination prevention.
M8: Multi-Agent Orchestration	Coordinated four specialized agents through a central orchestrator with routing logic, validation, and fallback mechanisms.

6. Detailed System Components

ClinicalBridge is built around five core components: four specialized LLM agents and a central orchestrator. Each agent performs a specific task within the workflow and is configured with different prompt strategies and temperature settings based on its responsibilities. This modular design improves reliability, simplifies prompt iteration, and allows each component to be evaluated independently before contributing to the final Clinical Context Brief.

Agent	Role	Temperature	Prompt Engineering Focus
Alert Triage	Classifies the urgency of incoming RPM alerts and generates targeted retrieval queries for downstream agents.	0.0	Deterministic prompt design, threshold-based classification, trend analysis, escalation rules, and structured JSON outputs.
EHR Retrieval	Retrieves relevant patient information from the EHR using Retrieval-Augmented Generation (RAG).	0.0	Grounded retrieval, hallucination prevention, source referencing, confidence scoring, and missing-data detection.
Anamnesis	Extracts and structures patient-reported symptoms, medication adherence, and	0.3	Interpretation of patient language, preservation of context, uncertainty handling, and structured information

Agent	Role	Temperature	Prompt Engineering Focus
	lifestyle information.		extraction.
Synthesis	Combines evidence from all agents to generate the Clinical Context Brief (CCB).	0.1	Cross-source reasoning, conflict detection, confidence calibration, evidence-based summarization, and clinician-focused reporting.
Orchestrator	Coordinates communication between agents and manages the overall workflow.	N/A	Workflow orchestration, schema validation, fallback handling, safety enforcement, and critical alert routing.

6.1 Clinical Context Brief Output

The Clinical Context Brief (CCB) is the final output generated by ClinicalBridge and serves as the primary interface between the system and the clinician. Its purpose is to present the most relevant patient information in a clear, structured format, allowing clinicians to quickly understand the patient's current condition without manually reviewing multiple data sources.

Each Clinical Context Brief includes an alert summary, a patient snapshot, contextual analysis, risk assessment, recommended clinician review actions, identified uncertainties or missing information, an overall confidence score, and a mandatory safety disclaimer. Every finding is supported by evidence retrieved from the EHR, RPM, or anamnesis data, helping ensure transparency and reducing the likelihood of unsupported conclusions. By organizing the information in a consistent structure, the CCB reduces the time required to review patient context while supporting informed clinical decision-making.

7. Data Model and Simulated Dataset

ClinicalBridge was developed and evaluated using a fully simulated clinical dataset consisting of 15 fictional adult patients. Five of these patients were specifically designed as evaluation scenarios, each representing a different clinical situation used to assess the performance of the system. The remaining ten patients were included to create a more realistic retrieval environment, ensuring that the EHR Retrieval Agent searched through multiple patient records instead of operating on a very limited dataset. This better reflects how retrieval systems perform in real-world clinical settings.

The dataset is divided into four components, each representing one part of the ClinicalBridge workflow.

Data Source	Format	Main Contents
EHR	JSON	Patient demographics, ICD-10 diagnoses, medication lists, allergies, laboratory results, clinical notes, and scenario metadata.
RPM	CSV	Time-series physiological measurements, including blood pressure, blood glucose, body weight, and oxygen saturation collected from simulated monitoring devices.
Anamnesis	JSON	Patient-reported symptoms, chief complaint, history of present illness, medication adherence, over-the-counter medication use, review of systems, social history, and symptom diary.
Gold standards	JSON	Reference Clinical Context Briefs (CCBs) used to evaluate system performance across the five clinical scenarios.

7.1 Scenario Coverage

Scenario	Patient	Capability Tested
1. The Missed Medication	PT001 - Mehmet Yildiz, 67M	Detects discontinued ACE inhibitor use and over-the-counter ibuprofen as contributing factors to elevated blood pressure.
2. The False Alarm	PT002 - Ayse Kaya, 54F	Uses EHR and anamnesis information to identify a glucose alert as non-critical due to a documented dietary change.
3. The Silent Deterioration	PT003 - Hasan Demir, 75M	Detects gradual heart failure deterioration through weight trends and identifies an NSAID allergy using information from multiple data sources.
4. The Incomplete Record	PT004 - Elif Sahin, 60F	Demonstrates the system's ability to operate with incomplete EHR records while extracting important information from anamnesis data.
5. The Conflicting	PT005 - Kemal	Identifies inconsistencies between reported medication adherence and supporting laboratory

Scenario	Patient	Capability Tested
Data	Aydin, 62M	evidence while maintaining neutral, non-accusatory language.

8. Prompt Engineering Specifications

Prompt engineering is the foundation of ClinicalBridge and plays a central role in the behavior of each agent. Rather than relying on a single prompt, the system uses specialized prompts tailored to the responsibilities of the Alert Triage Agent, EHR Retrieval Agent, Anamnesis Agent, and Synthesis Agent. Throughout development, each prompt was iteratively refined based on testing results, observed failure cases, and evaluation across the predefined clinical scenarios.

The prompt engineering process focused on improving reliability, reducing hallucinations, enforcing structured outputs, and ensuring that every generated response remained grounded in the available clinical data. Particular attention was given to defining clear agent responsibilities and limitations so that no agent generated diagnoses or treatment recommendations beyond the scope of the project.

Design Principle	Implementation
Role + boundary	Each agent is assigned a clearly defined responsibility along with explicit instructions describing its limitations and prohibited behavior.
Structured outputs	All agents generate structured JSON outputs based on predefined schemas to ensure consistent communication between components.
Few-shot and negative examples	Prompt examples demonstrate both correct behavior and responses that should be avoided, particularly those involving hallucinations or unsupported clinical conclusions.
Temperature tuning	Temperature values were selected according to each agent's role: 0.0 for deterministic retrieval and classification, 0.3 for patient language interpretation, and 0.1 for synthesis.
Confidence calibration	Confidence scores are assigned based on evidence quality, retrieval results, and consistency across multiple data sources.
Safety redundancy	Prompt guardrails, mandatory disclaimers, allergy detection, and no-diagnosis rules are reinforced throughout the workflow by both the

Design Principle	Implementation
	individual agents and the orchestrator.

8.1 Prompt Iteration Summary

Agent	v1 to v2 Improvement	v2 to v3 Improvement
Alert Triage	Added cumulative heart failure weight-gain thresholds, escalation modifiers, dynamic evaluation windows, and paired-measurement handling.	Improved handling of benign alerts by introducing provisional informational classifications and contextual guidance.
EHR Retrieval	Added pattern recognition, stale-record detection, and confidence scoring.	Improved pattern detection and refined retrieval trigger conditions.
Anamnesis	Added caregiver identification, defensive-language detection, and standardized date extraction.	Improved extraction of previous hospitalizations and patient history.
Synthesis	Added allergy verification, confidence scoring, and improved output ordering.	Prevented unsupported clinical reference values and introduced mandatory recommendations when EHR confidence is low.

9. Evaluation Framework and Results

The performance of ClinicalBridge was evaluated using both agent-level metrics and end-to-end system evaluation across the five predefined clinical scenarios. The evaluation framework was designed to measure retrieval accuracy, prompt reliability, safety compliance, and the overall quality of the generated Clinical Context Briefs (CCBs).

The initial (v2) evaluation served as the baseline for identifying areas that required improvement. Based on the observed results, several prompt refinements were introduced, leading to the final implementation. After these revisions, the evaluation confirmed that the system successfully completed all five clinical scenarios and passed all 40 individual validation checks, demonstrating improved reliability and consistent performance across the complete workflow.

9.1 Agent-Level Metrics

To better understand the performance of each component, every agent was evaluated independently using task-specific metrics. These measurements assess how accurately each agent performs its assigned role before contributing to the overall Clinical Context Brief. The results also helped identify areas requiring prompt refinement during development, particularly for the Alert Triage Agent, while confirming strong performance across retrieval, synthesis, and safety-related tasks.

Agent	Metric	Target	v2 Result	Status
Triage	Urgency accuracy	$\geq 90\%$	80% (4/5)	Revised in v3
Triage	Query relevance	$\geq 4/5$	4.4/5	Pass
EHR	Retrieval precision	$\geq 80\%$	84%	Pass
EHR	Retrieval recall	$\geq 75\%$	81%	Pass
EHR	Allergy detection	100%	100%	Pass
Anamnesis	Extraction completeness	$\geq 85\%$	89%	Pass
Anamnesis	Interpretation accuracy	$\geq 80\%$	87%	Pass
Anamnesis	Adherence confidence	100%	100%	Pass
Synthesis	Clinical accuracy	$\geq 90\%$	92%	Pass
Synthesis	Hallucination rate	$\leq 5\%$	3.2%	Pass
Synthesis	Completeness	$\geq 85\%$	88%	Pass
Synthesis	Source traceability	$\geq 90\%$	91%	Pass

The table presents the v2 evaluation results because they served as the baseline used to identify weaknesses during prompt iteration. The improvements introduced in v3 are summarized in the following sections and reflected in the final end-to-end evaluation.

9.2 End-to-End Metrics

The end-to-end evaluation measures the overall performance of ClinicalBridge after all agents are integrated into a single workflow. These metrics assess how effectively the complete system processes an RPM alert, retrieves relevant clinical information, generates a Clinical Context Brief, and satisfies the project's safety and reliability requirements.

Metric	Target	v2 Result	Final Evaluation
Scenario pass rate	5/5	4/5	5/5
Time-to-brief	<30 seconds	23.4 seconds avg	N/A in mock mode
Safety compliance	100%	100%	40/40 checks include disclaimer/no diagnostic language
Source traceability	>=90%	91%	All scenarios passed source citation validation

9.3 Failure Analysis and Improvements

The initial evaluation highlighted several areas where prompt design could be improved. Rather than modifying the system architecture, these issues were addressed through targeted prompt refinements and updated decision rules. The table below summarizes the main observations, their underlying causes, and the improvements introduced in the final version of the system.

Observed Issue	Root Cause	Improvement Implemented (v3)
Scenario 2 triage: Routine vs Informational	The Alert Triage Agent lacked sufficient contextual guidance for diet-related alerts.	Added provisional informational criteria and contextual reclassification rules.
Scenario 4 synthesis:	Low EHR confidence was detected but was not converted	Added an automatic records-request recommendation

Observed Issue	Root Cause	Improvement Implemented (v3)
records request omitted	into a recommended action.	whenever EHR confidence falls below 0.5.
Unsourced clinical reference values	The Synthesis Agent occasionally included background reference values that were not supported by retrieved evidence.	Restricted the model to using only values explicitly available in the retrieved agent outputs.
Previous hospitalization not extracted	Hospitalization history remained in free-text patient narratives and was not captured as structured information.	Added a dedicated extraction rule for previous hospitalization mentions.

10. Project Milestones and Timeline

Week	Milestone	Completed Evidence
1	Foundation and design	Completed problem analysis, selected the LLM, designed the system architecture, created the dataset schema, and prepared the initial simulated patient records.
2	Prompt engineering and RAG	Developed the first prompt library (v1), designed the Clinical Context Brief (CCB) schema, implemented the ChromaDB retrieval pipeline, and documented the workflow.
3	Agent implementation	Implemented the four specialized agents, completed unit testing, performed the initial evaluation, analyzed failure cases, and refined prompts to version 2.
4	Multi-agent integration	Integrated the orchestrator, implemented safety guardrails, added fallback mechanisms and critical alert handling, and finalized prompt version 3.
5	Evaluation and portfolio	Completed the evaluation report, prompt engineering portfolio, demonstration notebook, final project report,

Week	Milestone	Completed Evidence
		and organized the repository for submission.

11. Deliverables Checklist

Deliverable	Project Location	Status
Project report	outputs/ClinicalBridge_Final_Project_Report.docx	Complete
Working prototype	main.py , agents/	Complete
Simulated dataset	data/ehr, data/rpm, data/anamnesis	Complete
Prompt engineering portfolio	docs/prompt_engineering_portfolio.md	Complete
Evaluation report	evaluation/evaluation_report.md + JSON outputs	Complete
Demonstration	demo/ClinicalBridge_Demo.ipynb	Complete
Repository README	README.md	Complete

12. Ethical Considerations and Limitations

ClinicalBridge was developed as an educational proof of concept to demonstrate the application of prompt engineering in a healthcare setting. Because the project operates in a safety-critical domain, ethical considerations were incorporated throughout the system design. The prototype was intentionally designed to support clinical decision-making rather than replace it, with multiple safeguards implemented to encourage transparency, reduce hallucinations, and ensure that all generated outputs remain grounded in the available evidence.

12.1 Critical Disclaimers

- ClinicalBridge is an educational prototype and is not intended for clinical use.

- All patient information used throughout the project is fully simulated. No real, de-identified, or confidential patient data was used during development or evaluation.
- The system does not generate medical diagnoses or treatment recommendations.
- Every Clinical Context Brief (CCB) should be reviewed and interpreted by a qualified healthcare professional before any clinical decisions are made.

12.2 Ethical Design Principles Implemented

The following principles guided the design and implementation of ClinicalBridge:

- **Transparency:** Every Clinical Context Brief includes confidence scores, source references, and uncertainty or missing-data indicators to support explainable outputs.
- **Human-in-the-Loop:** The system is designed to assist clinicians by presenting relevant information while leaving all medical decisions to qualified healthcare professionals.
- **Safety-First Design:** Critical alerts can be prioritized and routed for immediate clinician review when appropriate.
- **Hallucination Reduction:** Prompts and retrieval mechanisms were designed to ensure that generated information remains grounded in the available EHR, RPM, and anamnesis data.
- **Bias Awareness:** The use of simulated data is acknowledged as a limitation, as it cannot fully represent the diversity and complexity of real clinical environments.

12.3 Known Limitations

Although ClinicalBridge demonstrates the potential of prompt engineering in healthcare, several limitations remain:

- The simulated dataset is relatively small compared with real clinical databases and does not capture the full complexity of hospital records.
- The reference Clinical Context Briefs were created by the project authors and may include subjective interpretations.
- Real Electronic Health Records often contain inconsistent formatting, abbreviations, scanned documents, and incomplete information that are not fully represented in the simulated dataset.
- The prototype has not undergone clinical validation and is not intended for deployment in real healthcare environments.
- Despite the implemented safety mechanisms, Large Language Models can still produce unsupported outputs, meaning the possibility of hallucinations cannot be completely eliminated.

12.4 Conclusion

ClinicalBridge demonstrates how prompt engineering can be applied to develop a context-aware, multi-agent clinical decision-support prototype. By integrating Electronic Health Records, Remote Patient Monitoring data, and patient anamnesis into a single Clinical Context Brief, the system provides clinicians with a structured overview of relevant patient information while maintaining transparency and safety. Although the project is limited to simulated clinical data, it demonstrates the potential of Retrieval-Augmented Generation, multi-agent orchestration, and carefully engineered prompts for supporting future healthcare applications.

13. References

13.1 Clinical Context Gap and Health Informatics

- [1] A. Wright *et al.*, "Problem list completeness in electronic health records: A multi-site study and assessment of success factors," *International Journal of Medical Informatics*, vol. 84, no. 10, pp. 784–790, 2015.
- [2] J. Adler-Milstein and A. K. Jha, "HITECH Act drove large gains in hospital electronic health record adoption," *Health Affairs*, vol. 36, no. 8, pp. 1416–1422, 2017.
- [3] M. Senbekov *et al.*, "The recent progress and applications of digital technologies in healthcare: A review," *International Journal of Environmental Research and Public Health*, vol. 17, no. 7, Art. no. 2431, 2020.

13.2 Remote Patient Monitoring

- [4] M. Cvach, "Monitor alarm fatigue: An integrative review," *Biomedical Instrumentation & Technology*, vol. 46, no. 4, pp. 268–277, 2012.
- [5] A. Vegesna *et al.*, "Remote patient monitoring via non-invasive digital technologies: A systematic review," *Telemedicine and e-Health*, vol. 23, no. 1, pp. 3–17, 2017.
- [6] B. Noah *et al.*, "Impact of remote patient monitoring on clinical outcomes: An updated meta-analysis of randomized controlled trials," *NPJ Digital Medicine*, vol. 1, no. 1, 2018.

13.3 LLM Applications and Prompt Engineering

- [7] K. Singhal *et al.*, "Large language models encode clinical knowledge," *Nature*, vol. 620, pp. 172–180, 2023.
- [8] H. Nori *et al.*, "Capabilities of GPT-4 on medical challenge problems," Microsoft Research, 2023.

- [9] A. J. Thirunavukarasu *et al.*, "Large language models in medicine," *Nature Medicine*, vol. 29, pp. 1930–1940, 2023.
- [10] J. Wei *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [11] S. Yao *et al.*, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations (ICLR)*, 2023.
- [12] J. S. Park *et al.*, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2023.
- [13] LangChain, "LangChain Documentation," Available: <https://python.langchain.com/>. Accessed: Jun. 2026.

Appendix A. Evidence and Verification Summary

This appendix summarizes the key project artifacts produced during the development of ClinicalBridge. Each item is linked to its corresponding location within the repository, providing traceability for the system design, prompt engineering process, implementation, and evaluation results.

Evidence	Source
Architecture and model selection	docs/architecture_design.md
Prompt iteration history	docs/prompt_engineering_portfolio.md and prompts/v1-v3/
Dataset design	data/DATASET_DOCUMENTATION.md
v2 measured evaluation	evaluation/evaluation_report.md and evaluation_results.json
Final mock evaluation	evaluation/evaluation_run_results.json; command output showed 5/5 scenarios and 40/40 checks passed
Prototype code	main.py, agents/, setup_vectorstore.py
Demo	demo/ClinicalBridge_Demo.ipynb

Verification Note: The final local evaluation confirmed successful completion of all five clinical scenarios and all 40 validation checks. While the project snapshot prevented a refreshed JSON results file from being written during the final execution, the console output verified that the implemented prompt refinements resolved the previously identified issues and that the complete evaluation pipeline executed successfully.